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01. Product Requirements Document (PRD)

Pura padhna ,mehenat lagi hai

**Kuch bhi add krna ho ya edit krna ho toh
group main discuss kr lena**

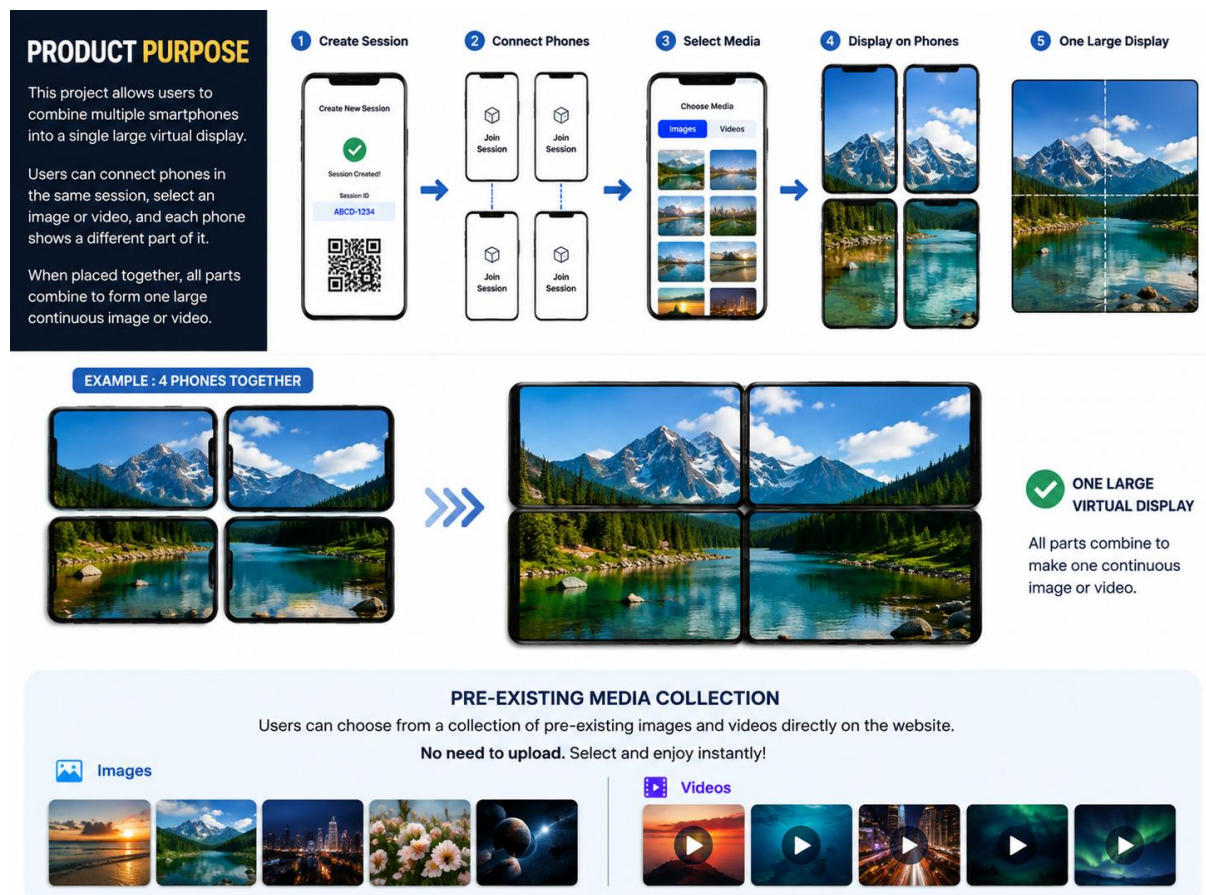
Product Requirements Document (PRD)

1. Product Purpose

The purpose of this project is to allow users to combine multiple smartphones into a **single large virtual display**.

A user can place multiple smartphones next to each other, connect them to the same session, select a local image or video, and display different portions of that media on each phone. When the phones are positioned together, the individual portions combine to create one large continuous image or video.

The website will also provide a collection of **pre-existing images and videos** that users can directly select and use without uploading their own media. This will allow users to quickly experiment with the virtual display and have fun creating large, continuous visuals across multiple smartphones.



2. Product Vision

The vision is to create a **simple, affordable, browser-based multi-screen platform** where multiple smartphones can work together as one large virtual display.

The user should be able to:

Open → Create Session → Connect Phones → Select Layout → Select Media → Play

The system should handle the splitting, positioning, synchronization, and display automatically as much as technically possible.

The platform is designed not only as a utility, but also as a **fun and social experience for friends and families**. When people gather together, they should be able to connect their smartphones and use them collectively to create a much larger screen, display fun content, play interactive experiences, and enjoy something together.

The website will include **pre-built interactive images, videos, animations, and experiences** so users can start having fun immediately without needing to upload their own content. Users can also use their own local images and videos.

A key vision is to turn this concept into an **online social trend**. Through platforms such as Instagram and other social media channels, the product can be marketed around the idea of:

“When you meet your friends or family, try this.”

The goal is to encourage people to record and share their experiences, making the multi-phone display itself a fun activity that people want to try during gatherings, parties, celebrations, and other social events.

For example, the website could provide a **virtual cake experience**. Multiple friends or family members can connect their phones to create one large cake across their combined screens. Once the cake is displayed, users can perform an interactive action such as **cutting the cake with a virtual knife**, triggering synchronized animations and effects across all connected phones. This type of

experience can be designed specifically to be visually engaging, memorable, and easy to record and share on social media.

The long-term vision is to build a platform where **connecting multiple phones becomes an activity in itself**—something people do together for entertainment, celebrations, challenges, games, creative experiences, and social-media-worthy moments.





Core Product Features

3.1 Create Session

A user can create a new multi-device display session.

The system generates:

- Unique Session ID
- QR Code
- Session joining link

The session creator becomes the ****Master Device****.

3.2 Join Session

Other smartphones can join the session by:

- Scanning a QR code
- Opening a joining link
- Entering a Session ID manually

After joining, the device becomes a ****Display Device****

5.1 Create Session

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Create Together See Bigger

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- ✓ Scanning a QR code
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- ✓ Entering a Session ID manually

After joining, the device becomes a **Display Device**.

Connect. Play. Create Memories. 🎉 👥 ⭐

Breathtaking Views Virtual Cake Party Fun & Games Create Together

🌐 No App Required 📱 Works on Any Phone ⚡ Real-Time Sync 👤 Perfect for Any Occasion 🚀 More Experiences Coming Soon

4.1 Device Management

The Master Device should be able to see all connected devices.

Example:

Connected Devices: 4

Device 1 ✓

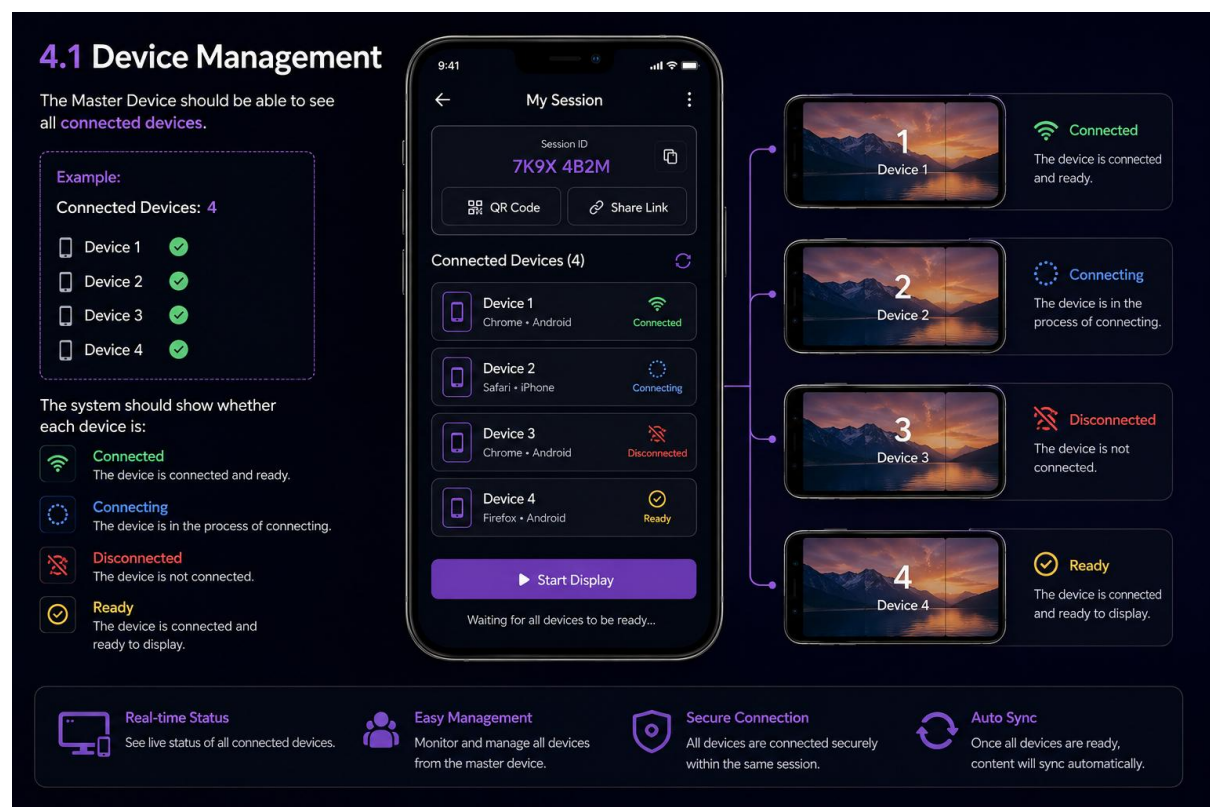
Device 2 ✓

Device 3 ✓

Device 4 ✓

The system should show whether each device is:

- Connected
- Connecting
- Disconnected
- Ready



5. Screen Layout Configuration

The user should be able to select how many screens are being used.

Initial supported layouts:

1 × 1

[1]

1 × 2

[1][2]

2 × 2

[1][2]

[3][4]

2 × 3

[1][2][3]

[4][5][6]

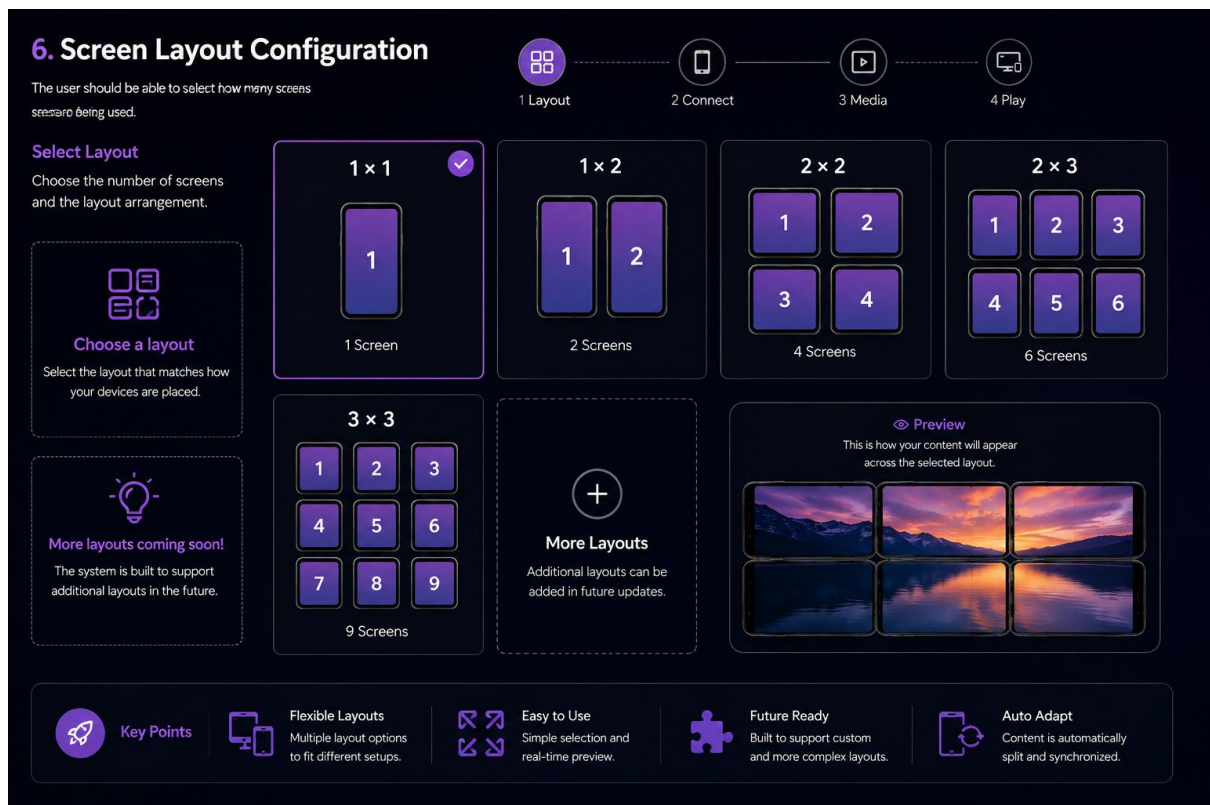
3 × 3

[1][2][3]

[4][5][6]

[7][8][9]

The system should be designed so additional layouts can be added in the future.



6. Device Position Assignment

When a user scans the **QR code** to join the session, the system will automatically assign a position to that phone based on the selected screen layout.

After scanning the QR code, an **animation will appear on the phone showing its exact position** in the overall screen layout.

For example:

Scan QR Code → Position Animation Appears → Check Your Position → Place Your Phone Accordingly

If the assigned position is **Top Right**, the phone will show an animation indicating **Top Right**, and the user simply needs to place their phone in that position next to the other connected phones.

The animation will make it easy for users to understand **where their phone needs to be placed**, without requiring complicated manual configuration.

Future Enhancement

In future versions, the system will be further upgraded to detect the **physical position of each connected phone**.

If a phone becomes slightly displaced from its assigned position, the system will detect the movement and inform the user, for example:

“Your phone is slightly out of position. Please move it to the correct position.”

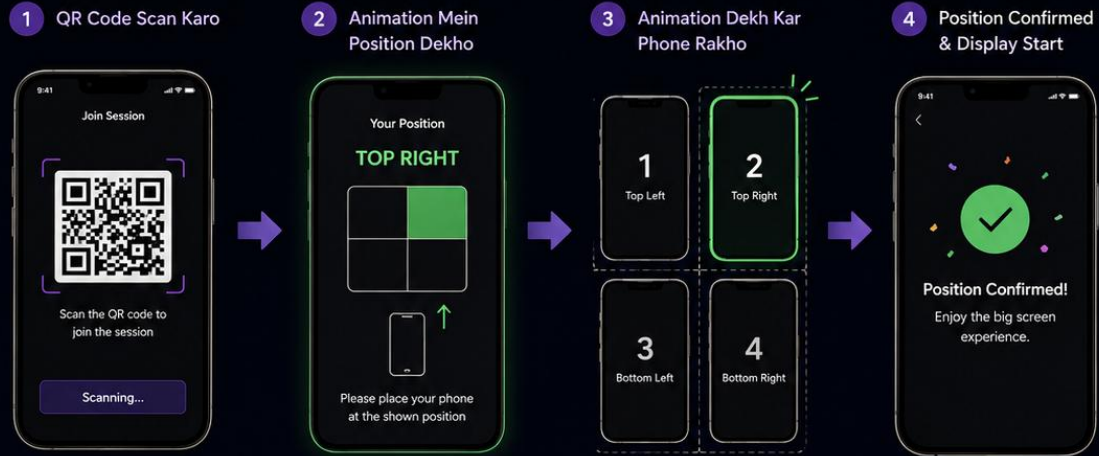
The system can eventually provide **real-time position detection and alignment guidance**, making it easier to maintain a perfectly aligned multi-phone display.

These advanced positioning and displacement-detection features are **planned for future versions and are not required for the MVP**.

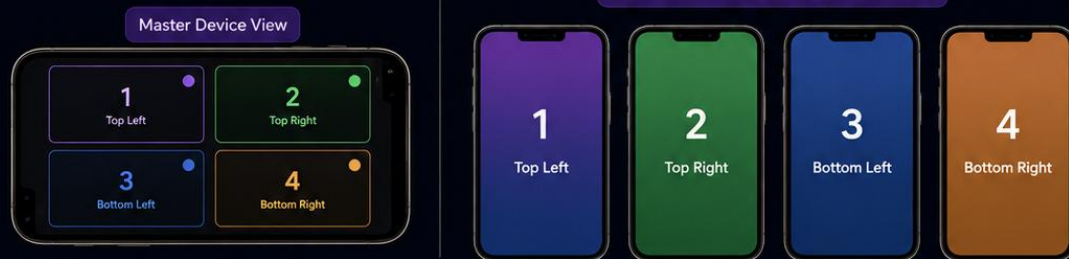
7. Device Position Assignment

Each connected phone must know which section of the media it needs to display.

How it Works (MVP)



Example: 2 x 2 Layout



Future Upgrade (Planned)

Future versions will detect if any phone is slightly displaced and help users correct it automatically.

- Smart Position Detection**
Detects even small displacement of any phone.
- Real-time Alerts**
Instantly notifies if a phone is not in the correct position.
- Easy Correction Guidance**
Shows where and how to adjust for perfect alignment.



We will keep improving the experience to make it smarter, easier and more fun!

7. Media Controls

The Master Device should control playback for all connected devices.

Required controls:

- ▶ Play

-  Pause
-  Resume
-  Seek
-  Restart
-  Volume control where applicable
-  Fullscreen

When the Master Device performs a playback action, the connected display devices should respond accordingly.

8. Bezel Adjustment

Physical smartphone bezels create small gaps between screens.

The application should eventually provide a **Bezel Adjustment** feature.

Users should be able to adjust:

- Horizontal offset
- Vertical offset
- Scaling
- Crop position

This allows the combined display to look more continuous.

9 User Flow

Master Device

Open Website



Create Session



QR Code Generated



Other Phones Join



Select Number of Screens



Assign Phone Positions



Select Image/Video



Prepare Media



Start Playback



All Phones Display Their Sections

Normal Device

Open Website



Scan QR Code



Join Session



Select/Receive Position



Wait for Media



Enter Fullscreen



Display Assigned Section

10 Functional Requirements

FR-01 — Session Creation

The system must allow a user to create a new session.

FR-02 — Session Joining

The system must allow other devices to join an existing session.

FR-03 — QR Pairing

The system should provide QR-based session joining.

FR-04 — Device Identification

The system must identify each connected device within the session.

FR-05 — Layout Selection

The system must allow the Master Device to select the display grid.FR-06 — Position Assignment

The system must allow each display device to receive an assigned grid position.

FR-07 — Image Selection

The system must allow the user to select a local image.

FR-08 — Video Selection

The system must allow the user to select a local video.

FR-09 — Media Splitting

The system must calculate which portion of the media belongs to each display device.

FR-10 — Media Rendering

Each device must render only its assigned section.

FR-11 — Playback Control

The Master Device must control playback across connected devices.

FR-12 — Synchronization

The system must attempt to keep video playback synchronized.

FR-13 — Fullscreen

Display devices must support fullscreen playback.

FR-14 — Device Disconnect Handling

The system must detect when a device disconnects.

FR-15 — Session Termination

The Master Device must be able to end a session.iske liye image genetare kr

Functional Requirements

Multi-Device Display System – Combine multiple smartphones into one large display to view images and videos as a single, seamless experience.

FR-01



Session Creation

The system must allow a user to create a new session.

FR-02



Session Joining

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FR-03



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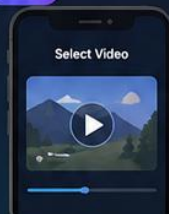
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FR-09



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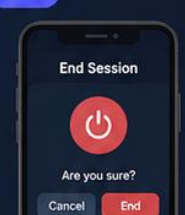
FR-14



Device Disconnect Handling

The system must detect when a device disconnects.

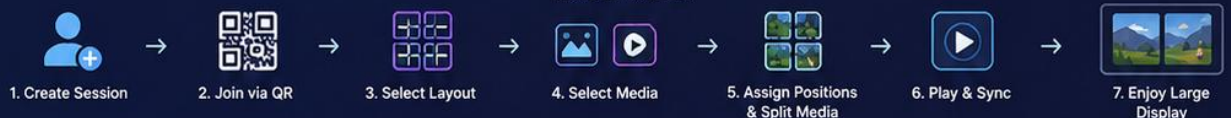
FR-15



Session Termination

The Master Device must be able to end a session.

How it Works



. 11 Non-Functional Requirements

Performance

The application should provide smooth media playback without unnecessary lag.

Responsiveness

The interface must work on:

- Smartphones
- Tablets
- Desktop browsers

Reliability

Temporary network interruptions should not crash the application.

Scalability

The architecture should allow additional display devices and layouts to be supported later.

Usability

A non-technical user should be able to create and operate a session without understanding the underlying technology.

Compatibility

The system should target modern browsers with support for:

- HTML5 Video
- Canvas
- WebSocket/WebRTC-related capabilities where required
- Fullscreen APIs

12. Future Features

Possible future versions may include:

- Automatic phone-position detection
- 3×3, 4×4 and larger layouts
- Better synchronization
- Automatic bezel correction
- Remote brightness control where supported
- Multiple media playlists
- Drag-and-drop screen positioning
- Cloud sessions
- PWA installation
- Offline/local-network mode
- Android native application
- iOS support
- Multi-user control
- Digital signage mode
- Scheduled media playback